

## TASK : 4

## Inheritance Based Application

Aim: To design and implement a simple Vehicle Rental System using inheritance in java, where different vehicle types calculate their rental cost based on a base rent.

### Algorithm:

1. Start
2. Create a base class Vehicle with the following data members:
  - \* vehicleId
  - \* modelName
  - \* base Rent
3. Define a constructor in the vehicle class to initialize the values.
4. Create a method display() in the base class to display vehicle details.
5. Create a derived class Car that inherits from Vehicle.
  - \* Add a method calculateRent() that add ₹500 to base rent.
6. Create another derived class Bike that inherits from Vehicle.
  - \* Add a method calculateRent() that adds ₹200 to base rent.
7. In the main() method:
  - \* Create an object of car and display its details and total rent.
  - \* Create an object of bike and display its details and total rent.
8. Stop.

### Program:

```
class Vehicle {
    int vehicleId;
    String modelName;
    double baseRent;
    Vehicle(int vehicleId, String modelName, double baseRent) {
        this.vehicleId = vehicleId;
        this.modelName = modelName;
        this.baseRent = baseRent;
    }
    void display() {
        System.out.println("Vehicle ID : " + vehicleId);
        System.out.println("Model Name : " + modelName);
        System.out.println("Base Rent : " + baseRent);
    }
}

class Car extends Vehicle {
    Car(int vehicleId, String modelName, double baseRent) {
        super(vehicleId, modelName, baseRent);
    }
    double calculateRent() {
        return baseRent + 500;
    }
}

class Bike extends Vehicle {
```



### Input:

#### Car Details:

Vehicle ID : 101

Model Name : Honda City.

Base Rent : 2000

#### Bike Details:

Vehicle ID : 201

Model Name : Yamaha R15

Base Rent : 800

### Output:

#### --- Car Details ---

Vehicle ID : 101

Model Name : Honda City.

Base Rent : 2000.0

Total Rent : 2500.0

#### --- Bike Details ---

Vehicle ID : 201

Model Name : Yamaha R15

Base Rent : 800.0

Total Rent : 1000.0

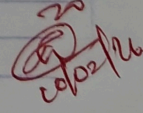


```

    Bike(int vehicleId, String modelName, double baseRent){
        super(vehicleId, modelName, baseRent);
    }
    double calculateRent(){
        return baseRent + 200;
    }
}

Public class Vehicle Rental {
    public static void main (String[] args){
        Car car = new Car (101, "Honda city", 2000);
        System.out.println(" --- Car Details --- ");
        car.display ();
        System.out.println (" Total Rent : " + car.calculateRent());
        System.out.println();
        Bike bike = new Bike (201, "Yamaha R15", 800);
        System.out.println(" --- Bike Details --- ");
        bike.display ();
        System.out.println(" Total Rent : " + bike.calculateRent());
    }
}

```

| VEL TECH - CSE          |                                                                                                   |
|-------------------------|---------------------------------------------------------------------------------------------------|
| EX NO.                  | 4                                                                                                 |
| PERFORMANCE (5)         | 5                                                                                                 |
| RESULT AND ANALYSIS (5) | 5                                                                                                 |
| VIVA VOCE (5)           | 5                                                                                                 |
| RECORD (5)              | 5                                                                                                 |
| TOTAL (20)              | 20                                                                                                |
| SIGN WITH DATE          | <br>20/02/26 |

Result: Thus, a Vehicle Rental System was successfully implemented using inheritance in Java, and the rental cost for car and bike was calculated correctly.